

SetProbe

Turning a language model's memory into complete database rows

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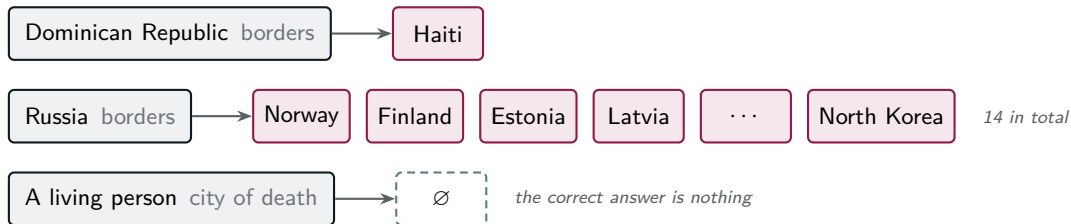
AKBC 2026 Shared Task

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Code: <https://github.com/JannicCutura/emnlp-kbc-2026>

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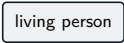
One question, three very different right answers

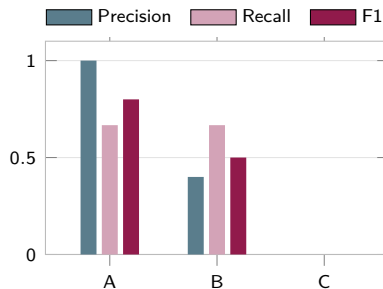


Given a subject s and a relation r , return the **complete** set of objects $\{o_1, \dots, o_k\}$ — closed-book, from the model's weights alone.

Why completeness is hard: every extra guess is scored

A country with three true neighbours. Three ways to answer it:

- A   *two right, one missed*
- B    *plus three wrong*
- C   *gold is ∅*



Three plausible extras cut precision from 1.0 to 0.4; one invented city turns a perfect row into a total miss. The question is not *whether* the model knows a fact but **when one more candidate should be admitted**.

The six relations

Relation	Object	Set size	Gold-empty
Area	number ($\pm 5\%$)	1	—
Capacity	number ($\pm 5\%$)	1	—
City of death	city	0–1	39%
Stock exchange	exchange	0–few	36%
Land borders	countries	0–14	26%
Awards	recipients	up to 610	—

Data

477 train, 478 validation, 475 test rows.
Validation gold is used for scoring only.

Score

Macro precision, recall and F1 over
subject–relation pairs, after normalisation
and alias expansion.

One relation wants a single number, another wants *nothing* a third of the time, a third wants hundreds of names. **Cardinality is the design problem.**

The rules: closed book

Allowed

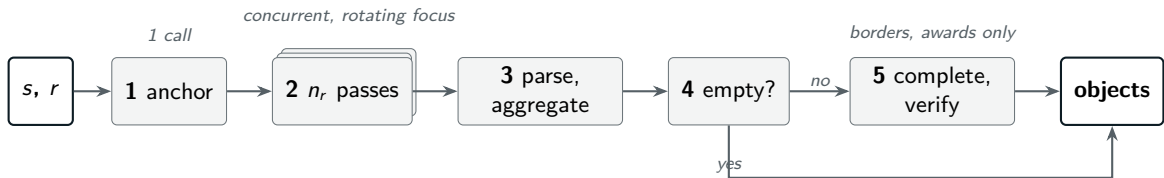
- ✓ Open-weight models, $\leq 32\text{B}$ published parameters
- ✓ In-context examples from the training split
- ✓ Multi-step inference, repeated sampling
- ✓ Non-neural post-processing

Forbidden

- ✗ Web, retrieval index, external corpus, knowledge base
- ✗ Fine-tuning
- ✗ Combining models whose parameters sum above 32B

Retrieval would supply exactly the long-tail facts a model's memory covers worst. Without it, **prompt diversity and repeated sampling** are the only sources of variation left.

One row of inference

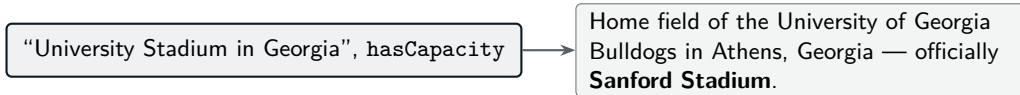


One open-weight model and one orchestration pattern for every relation. The next seven slides walk through each box and what it changed.

Step 1 — Say who the subject is before answering

Many subjects are ambiguous names. One cheap call (256 tokens, low temperature) asks the model to describe the exact entity first; that description is handed to every later pass as *uncertain* context.

anchor



- Resolves namesakes, scope, and “is this company independently listed?”
- It can also propagate a wrong guess: *Ayon Island* (East Siberian Sea) was anchored to the Philippines and every sample followed.
- **Not ablated**: present in every run, so its net value is unmeasured.

Step 2 — Ask many times, each time differently

Each pass uses a different seed, a different *recall focus*, and a different rotating set of training examples. Passes run concurrently.



- Every pass returns a short rationale and a JSON answer array.
- A pass the parser cannot read is dropped, never counted as “no objects” — it never happened in 5,392 validation calls.
- Aliases seen in training are canonicalised *before* votes are counted, so equivalent surfaces reinforce rather than split a vote.

Step 3 — Aggregate by cardinality, not by relation name

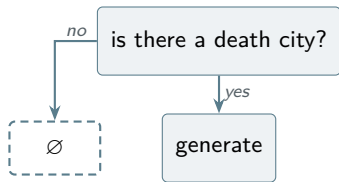
Relation	n_r	Rule
Area, capacity	7	median
City of death	20	plurality vote, <i>empty counts</i>
Stock exchange	10	support ≥ 0.30 , empty vote
Land borders	3	union
Awards	20	support ≥ 0.05 union

- **One value:** agreement is the signal $\rightarrow$ median.
- **Zero or one:** silence is a vote $\rightarrow$ plurality.
- **Many:** a name seen once in twenty passes may still be right $\rightarrow$ low-threshold union.

All thresholds were selected on **training traces only**; validation gold is never touched before scoring.

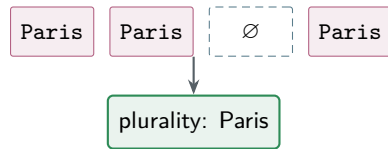
Step 4 — Let “nothing” be an answer, not a gate

Before: gate, then generate



A classifier suppressed **28 of 61** real death cities before the model was ever asked for one.

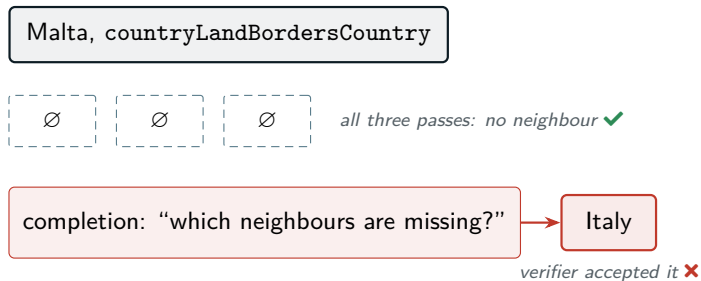
After: generate, then vote



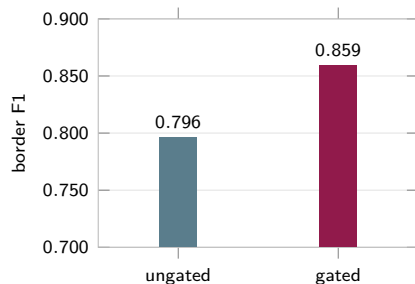
Every pass may return an empty set; empties vote like any other answer (threshold 0.5).

Also: few-shot demonstrations now mirror each relation's *own* empty rate (2 of 5 for death cities, 1 of 5 for borders). A fixed 60% empty mix had nudged the model toward silence on exactly the relations where silence is scored.

Step 5 — Never ask “what is missing?” of an empty set



Across the split the omission pass *added* 36 neighbours and verification removed 22. A leading question presupposes a set to have omissions from.



Post-processing now runs only on a non-empty aggregate.

Step 6 — Enumerate big sets in slices

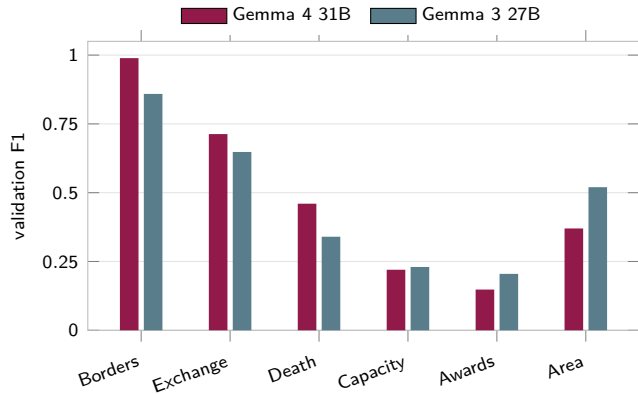
A single award may have 610 recipients; one undifferentiated prompt invites both truncation and omission.

20 passes cycle through six foci; union with support ≥ 0.05



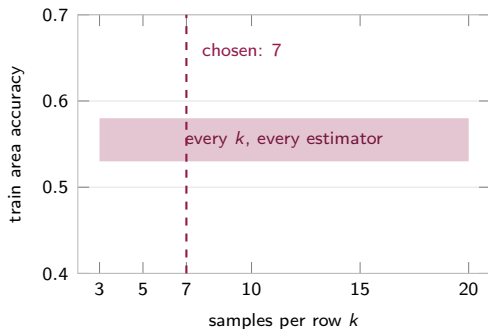
- Token cap raised 2,048 $\rightarrow$ 4,096: a truncated JSON array parsed as *zero* recipients, not a partial list.
- Mean predicted set size: 15 $\rightarrow$ 68 (gold mean 146).
- An earlier binary verifier cut correct recipients from 96 to 56, so we removed it: a closed-book verifier consults the same weights as the generator.

Step 7 — Choose the model, and pin how it is served



- Complementary, not ordered:
Gemma 4 wins where a precise definition matters, Gemma 3 where numeric recall matters.
- Overall 0.513 vs 0.490 — **not significant** (CI $[-0.02, +0.07]$).
- The best mix would score 0.544 but sums to 58.1B parameters.
- A hosted gateway serves one model name at anything from fp4 to bf16, so we pin provider and precision on every call.

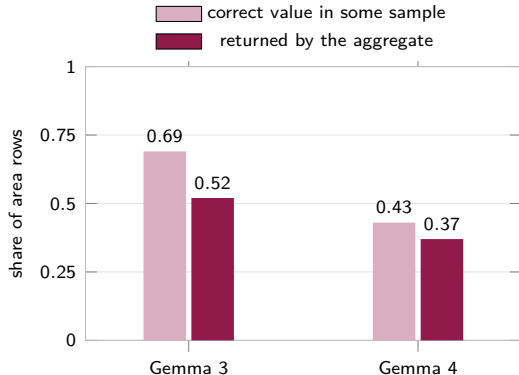
Three knobs that did nothing



- ❶ **Sample count.** Flat from 3 to 20 passes; twenty cost three times as much as seven and bought nothing.
- ❷ **Aggregation rule.** Mode-seeking, trimmed and magnitude-filtered estimators all tied or lost to the median.
- ❸ **Abstention thresholds.** Sweeping them on training traces rediscovered the defaults exactly.

None of the four factors — these three, or the model — moves the overall score beyond its 0.045 confidence interval.

Why: no rule selects a value that was never generated



- The light bar is the **ceiling**: the best any selection rule could do on that candidate pool.
- The gap between the bars is all that aggregation can be blamed for. The gap above the light bar is knowledge the model never produced.
- Of the incorrect rows, 58 of 63 (area) and 71 of 78 (capacity) contain no correct value in *any* response.

Validation results

Relation	G4 31B	G3 27B	Δ
Land borders	.989	.859	+.130*
Stock exchange	.713	.648	+.064
City of death	.460	.340	+.120
Capacity	.220	.230	-.010
Awards	.148	.205	-.057
Area	.370	.520	-.150*
All relations	.513	.490	+.022

* 95% paired-bootstrap interval excludes zero.

Headline

Macro F1 **0.513** on 478 validation rows, 95% CI [0.468, 0.557]; 0.484 on an independent training-split run; 0.460 for our earlier gated prototype.

Test split

Gold is not distributed. Predictions were frozen on 8 Aug 2026 under the locked configuration and await the organisers' score.

Where the remaining errors live: too quiet, not too loud

Stock exchange

36 of 36 gold-empty rows
correctly returned $\emptyset$ ✓

26 real listings
returned $\emptyset$ anyway ✗

City of death

33 of 39 gold-empty rows
correctly returned $\emptyset$ ✓

32 recoverable cities
suppressed ✗

- Replacing the hard gate with a vote changed the *mechanism*, not the *operating point*: when the model does not know, twenty passes independently return nothing, and no rule over unanimous abstentions manufactures a city.
- Exchanges are different: in 9 of 27 zero-score rows the rationale names the right exchange, then returns $\emptyset$ because it believes the listing has ended. That ceiling is temporal belief, not missing knowledge.

- ① **Sampling and aggregation are not where the score is.** Sample count, aggregation rule, abstention threshold, and even the choice between two eligible models all stay inside the confidence interval overall. The model alone moves *individual* relations, in opposite directions.
- ② **Abstention is a prediction.** Let empty answers vote; never gate generation, and never ask an empty set what it is missing.
- ③ **Report the ceiling with the score.** The share of rows whose correct value appears in *any* sample tells you whether to spend effort on elicitation or on aggregation. For us the answer was elicitation.

Code, prompts, traces and configuration:
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